

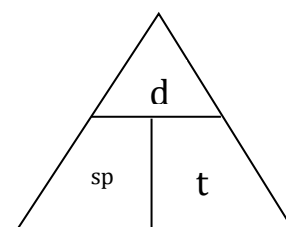
**YEAR 10 SCIENCE
EARTH AND SPACE SCIENCES
REVISION**

1. Complete the following table.

TERM	DEFINITION
light year	
constellation	
galaxy	
nebula	
nova	
protostar	
red giant	
neutron star	
black hole	
black dwarf	

2. Light travels at 300,000 km/sec. A star is 65.0 light years away.

- (a) How many seconds are there in 65.0 years?
(365 days x 24 hours x 60 mins x 60 secs)
- (b) Calculate how many kilometres are in 65.0 light years.
(distance = speed x time)



- (c) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer - a calculation of time would be very good.

(Assume the fastest rockets can travel at 100,000 km/h - about 28 km/sec.)

YES / NO (Circle your answer.)

Reasons:

-
-
-

3. (a) Name the process responsible for producing the energy released from the Sun.

- (b) What is the major element found in the Sun?

- (c) Describe how this element is involved in producing energy.

- (d) The Sun loses 5 million tonnes (5.00×10^9 kg) of mass per second. According to Einstein's equation ($E = mc^2$), how much energy (in Joules - J) is released?

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
cool	
moderate	
hot	

5. (a) About how old is our Sun? _____

- (b) How much longer will it "live"? _____

6. Describe how does a nebula become a protostar?

7. What is the major factor that determines the pathway a star takes in its life once it has formed?

8. (a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → → →

- (b) Do similar flow diagrams for the following types of stars.

Huge star → → →

Giant star → →

- (c) What is the difference between a ***nova*** and a ***supernova***?

Nova:

Supernova:

- (d) What is the difference between a ***black dwarf*** and a ***black hole***?

Black dwarf:

Black hole:

9. Why do the stars appear to shift slightly in the night sky throughout the year?
10. How did the observation of the stellar parallax effect change our ideas about the universe?
11. All of the stars in the constellation Orion appear to be at the same distance from Earth. Explain how observers on Earth know they are not.

12. Why are most stars found on the main sequence of the Hertzsprung-Russell diagram?
13. Why are there dark lines on the spectrum of light emitted by stars?
14. (a) Which colour - red or blue - has the highest frequency?
- (b) What is a red shift? What does it tell us about how a star or galaxy is moving relative to Earth.
- (c) What did Edwin Hubble find out about the movement of distant galaxies?

15. (a) How old is the universe thought to be?
- (b) According to the Big Bang Theory, what was present at the beginning of the universe?
- (c) Give a brief description of how the Big Bang Theory describes the formation of the universe.
- (d) The Steady State Theory was in competition to the Big Bang Theory. Briefly describe this theory.
- (e) How did the Steady State Theory explain that the universe was expanding, yet remained the same?
- (f) What evidence put an end to the Steady State Theory?

(g) Describe 3 major pieces of evidence that supported the Big Bang Theory.